

Epinephrine

- I. **Classification**
 - Sympathomimetic agent
- II. **Action(s)**
 - Stimulates alpha and beta receptors
 - Causes smooth muscle relaxation in the airways, increases cardiac output, and contraction of the smooth muscle that lines most arterioles.
- III. **Indication(s)**
 - Cardiac Arrest
 - Hypotension or Post-ROSC Care
 - Anaphylaxis
 - Bradycardia
 - Severe stridor / croup
 - Severe Asthma or COPD
- IV. **Contraindications**
 - Coronary insufficiency
- V. **Precautions**
 - HTN
- VI. **Drug interactions**
 - Beta blockers will inhibit effects
 - Haldol may decrease or reverse effects
 - Tricyclic antidepressants may cause prolonged hypertension
- VII. **Adult Dosing and Administration**
 - Cardiac Arrest – 0.5 mg IV/IO (1:10,000), dose **every 5 minutes**
 - VF / Pulseless VT
 - PEA / Asystole
 - Hypotension
 - Epinephrine Push Dose (1:100,000) 10-20 mcg (1-2 mL) every 2-3 minutes to maintain SBP >90 mmHg
 - Max total 100 mcg / 10 mL
 - Consult Medical Control for further doses
 - Epinephrine Drip start at 0.1 mcg/kg/min (range 0.05 – 5 mcg/kg/min) to maintain SBP >90mmHg
 - Bradycardia
 - Epinephrine Push Dose (1:100,000) 10-20 mcg (1-2 mL) every 2-3 minutes
 - Max total 100 mcg / 10 mL
 - Consult Medical Control for further doses
 - Epinephrine Drip start at 0.1 mcg/kg/min (range 0.05 – 5 mcg/kg/min) to maintain SBP >90mmHg
 - Anaphylaxis
 - Epinephrine Autoinjector 0.3mg IM, may repeat every 5 min x 3 doses max
 - Epinephrine 0.5 mg IM (1:1000), may repeat every 5 min x 3 doses max
 - Epinephrine Push Dose (1:100,000) 10-20 mcg (1-2 mL) every 2-3 minutes to maintain SBP >90 mmHg
 - Max total 100 mcg / 10 mL
 - Consult Medical Control for further doses
 - If maxed IM Epinephrine, Epinephrine Drip start at 0.1 mcg/kg/min (range 0.05 – 5 mcg/kg/min) to maintain SBP >90mmHg

- Severe Asthma / COPD –
 - Epinephrine Nebulize 3mg (3 vials 1:1,000)
 - Epinephrine 0.5mg IM (or autoinjector dose of 0.3mg IM)

VIII. Pediatric Dosing and Administration

- Cardiac Arrest – 0.01 mg/kg IV/IO, dose **every 5 minutes**
 - Neonatal Resuscitation – 0.03 mg/kg IV/IO every 3-5 minutes
- Hypotension
 - Epinephrine Push Dose 5-10 mcg (0.5-1 mL) IV/IO every 2-3 min to maintain SBP >90 mmHg or age based normal
 - Age >8 years Or Call Medical Director for Orders
 - Max total 100 mcg / 10 mL
 - Epinephrine drip start 0.1mcg/kg/min (range 0.05 – 5 mcg/kg/min) IV/IO to maintain SBP >90 mmHg or age based normal
- Bradycardia
 - *IF CPR*: 0.01 mg/kg (1:10,000)(max 1 mg) IV/IO every 5 minutes
 - Push Dose (1:100,000) 5-10 mcg (0.5 – 1 mL) IV/IO every 2-3 min (Age >8 years Or Call Medical Director for Orders)
- Anaphylaxis
 - Epinephrine 0.01 mg/kg (1:1,000) IM (max dose 0.5 mg), may repeat every 5 min to max 3 doses
 - Epinephrine Push Dose pressor IV/IO (1:100,000) 5-10 mcg (0.5mL – 1mL), repeat every 3 mins PRN
 - Epinephrine infusion 1mg in 1000ml NS infuse at 0.1mcg/kg/min. Titrate every 3 min to MAP 65 (max 10mcg/min)
- Severe Asthma – Nebulize 3mg (3 vials 1:1,000)
 - 0.01 mg/kg (1:1,000) IM (max dose 0.5 mg)
(Requires Medical Director Consult)
- Severe stridor/croup – Nebulize 3mg (3 vials 1:1,000)

IX. Onset

- 1-2 minutes

X. Duration

- 5-15 minutes

XI. Possible Adverse Effects

- Cardiac arrhythmias
- Hypertension
- Chest Pain / Myocardial Infarction / Tachycardia
- Anxiety / Headache

XII. Other Notes

- **Epinephrine Push Pressor**
 1. Start with Epi 1:1,000 and transfer 1 mL to 100 mL NS
OR
 1. Start with Epi 1:10,000 and transfer 1 mL to empty 10 mL syringe
 2. Add 9 mL NS to the syringe
- **Epinephrine Drip**
 1. Add 2 mg (2 mL) of Epi 1:1,000 to NS 500 mL